
Introduction

The most important medium through which we learn in transdisciplinary education is the complex problem.

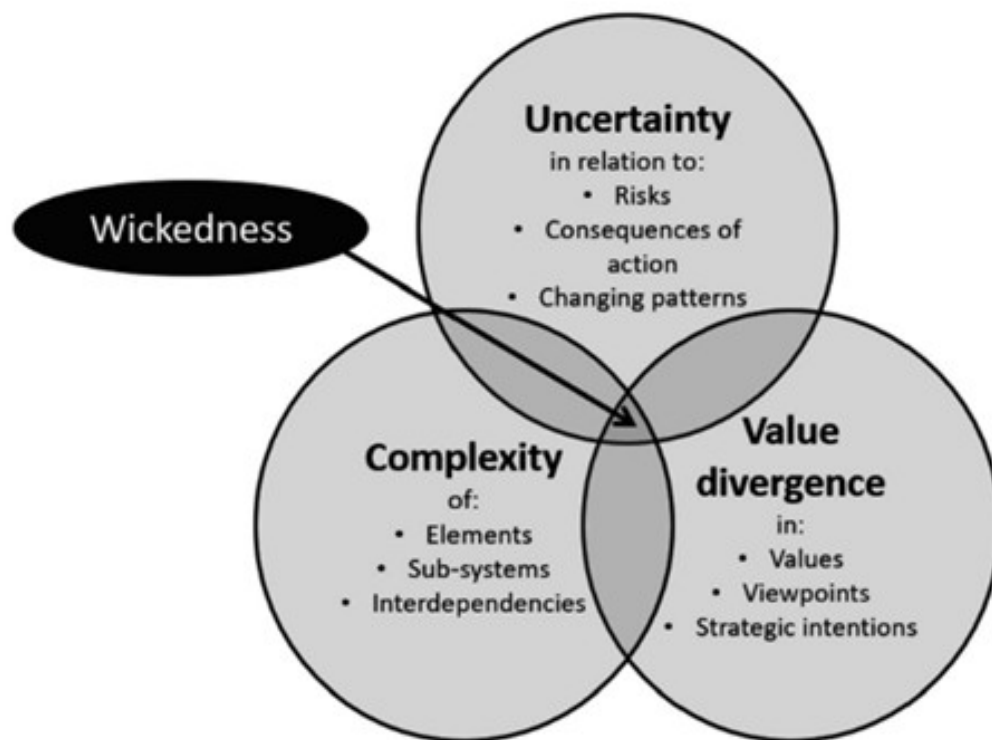
The **Theory** page explores the characteristics of such a challenge and how it influences your educational design.

Under the **Practice** tab, you'll find the exercise "From societal challenge to learnable task", which can help you make this crucial translation.

Theory

Complex problems, also known as wicked problems, have several defining characteristics:

1. Uniqueness: Each problem is unique and cannot be solved by standard methods.
2. Interconnectedness: Problems are interconnected and cannot be solved in isolation.
3. Uncertainty: The boundaries of the problem are unclear and change over time.



Schematic visualization of a wicked problem (Veltman et al., 2021)

When designing transdisciplinary education, it's important that the complex problem meets these characteristics.

At the same time, the challenge must also be learnable — meaning it should be structured in a way that allows participants to learn from it.

A well-formulated challenge gives students room to explore and investigate, while also providing direction.

By carefully framing challenges, you create a learning environment in which students actively contribute to societal change and deepen their own learning process.

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Learnability Aspect	Reflection Questions
Concrete problem with insight into causes	What is the problem? What lies beneath the surface?
Specific context	Where is this happening? What is characteristic of this place or organization?
Multiple stakeholders	Who is involved? Which interests or perspectives are in conflict?
Direction without fixed answers	What solution direction is worth exploring, without already defining a solution?

Reference:

Veltman, M. E., Van Keulen, J., & Voogt, J. M. (2021). Using problems with wicked tendencies as vehicles for learning in higher professional education: Towards coherent curriculum design. *The Curriculum Journal*, 32(3), 559-583.

Practice

Exercise: From Societal Issue to Learnable Challenge

Based on the societal issue “*Diversity in Social Housing*” from the **Stadsacademie Gent** Block, T., Prové, C., Dehaene, M. et al. *Understanding urban sustainability from Mode 2 Science and transdisciplinary education: how Master Thesis Ateliers of the Ghent Stadsacademie tackle wicked issues. Environment, Development and Sustainability* (2022).

<https://doi.org/10.1007/s10668-022-02657-0>

Step 1 – Choose a Concrete Societal Issue

Select a current issue from a region, neighborhood, or organization. Choose something that is complex, has multiple perspectives, and is socially relevant.

Tip: Think of themes that connect to your educational practice or to partners in the field.

Example:

Access to social housing for people with a migration background in Ghent is difficult, despite formal equality in the regulations.

Step 2 – Analyze the Issue Using Four Learnability Aspects

Answer the questions below to deepen your understanding of the issue. Write your answers briefly and concretely.

Step 3 – Formulate a Learnable Challenge

Use an open-ended formulation that invites research or design. For example:

- How can...
- What is needed to...
- In what way can...

Example:

How can the allocation system for social housing in Ghent better reflect the diversity of prospective tenants, especially people with a migration background, given the tensions between formal equality, selection criteria, and perceived inequality?

Step 4 – Check: Is It All There?

Verify whether your formulated challenge meets the four learnability aspects. (in theory section)